

Experiment#02

Objective:

In this experiment, we will study and implement the following gates (AND, OR, NOT).

Equipment's:

- Breadboard
- Power supply
- Jumper wire
- Integrated (IC's)
- Led light

Theory:

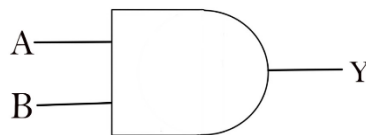
And-gate:

An AND gate is a fundamental logic gate in digital electronics that performs the logical AND operation on its inputs. In simpler terms, it acts like a gatekeeper, only allowing the output to pass through if all of its inputs are true (represented by the value 1). If even one input is false (represented by the value 0), the output remains blocked (at 0).

Expression:

$$Y=A.B$$

Symbol:



Fig#01

OR -gate:

An OR gate is a fundamental logic gate that performs the logical OR operation. In simple terms, it outputs a HIGH (true) signal if at least one of its inputs is HIGH, and a LOW (false) signal only if both inputs are LOW.

Expression:

$$Y=A+B$$

Symbol:



Fig#02

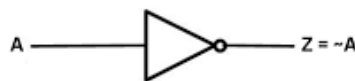
Not-gate:

A NOT gate, sometimes called an inverter, is a fundamental building block in digital logic circuits. It's a simple yet powerful logic gate with just one input and one output. Its sole purpose is to flip the input signal, meaning:

- If the input is HIGH (1), the output becomes LOW (0).
- If the input is LOW (0), the output becomes HIGH (1).

Expression:

$$Y = \neg A$$

Symbol:

Fig#03

Procedure:**AND-gate**

- First, collect all the required equipment and components that are used.
- Now put 7408 IC on the breadboard
- Then, put the input supply connection to the IC with jumper wires that are connected at PIN 14 for positive input and at PIN 7 for ground/negative supply.
- Now give input supply from PIN 14 and 7 to PIN 1 and 2 as input that invert is given on the output side.
- For checking, put the LED light on the output side.
- After all the connections have been done. Then check all the connections carefully.
- Now give the power supply to the circuit with the help of the DC power supply.

OR gate:

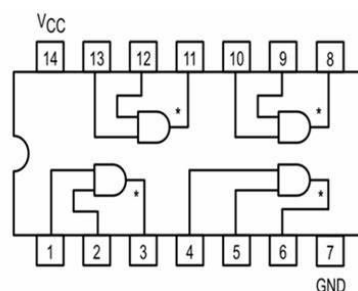
- First, collect all the required equipment and components that are used.
- Now put 7432 IC on the breadboard
- Then, put the input supply connection to the IC with jumper wires that are connected at PIN 14 for positive input and at PIN 7 for ground/negative supply.
- Now give input supply from PIN 14 and 7 to PIN 1 and 2 as input that invert is given on the output side.
- For checking, put the LED light on the output side.
- After all the connections have been done. Then check all the connections carefully.
- Now give the power supply to the circuit with the help of the DC power supply.

Not gate:

- First, collect all the required equipment and components that are used.
- Now put 7404 IC on the breadboard
- Then, put the input supply connection to the IC with jumper wires that are connected at PIN 14 for positive input and at PIN 7 for ground/negative supply.
- Now give input supply from PIN 14 to PIN 1 as input that invert is given on the output side. Because NOT gate has only one input pin that inserts a
- For checking, put the LED light on the output side.
- After all the connections have been done. Then check all the connections carefully.
- Now give the power supply to the circuit with the help of the DC power supply.

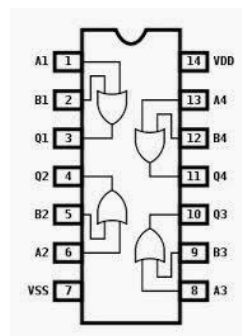
Circuit Diagram:

AND gate:



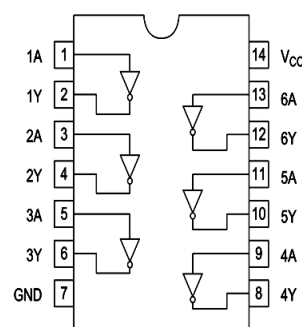
Fig#04(AND gate)

OR gate:



Fig#05(OR gate)

NOT gate:



Fig#06(NOT gate)

Truth table:

And gate:

A	B	$Y=A.B$

OR gate:

A	B	$Y=A+B$

NOT gate:

A	$Y=\neg A$

Conclusion:

In this experiment, we have learned about different types of gates working, truth tables, and equations. It also helps to know how to make circuit diagrams and how we make circuits with the help of circuit diagrams and ICs on breadboard.